Introduction

Introduction I

Rapidly changing fields require literature reviews that preserve the distinction between what was searched, what was retrieved, what was assessed, and what can be claimed. A single broad query hides phase-specific coverage and makes it difficult to explain how records entered a synthesis. This project treats a literature review as a reproducible data pipeline: configuration defines the review design, source adapters acquire records, analysis modules produce evidence artifacts, and rendering turns those artifacts into an auditable manuscript.

The reporting surface follows the spirit of systematic-review transparency (Page et al. 2021) without implying that the bundled fixture satisfies a domain protocol. A live review must define eligibility criteria, search dates, source coverage, screening procedures, and a domain-appropriate synthesis plan before its results are interpreted.

The bundled configuration targets **Exoplanet Atmospheres**. It defines three ordered phases for foundational methods, JWST-era observations, and molecular detection. The phase labels are configuration, not conclusions: a live review must justify its boundaries and update its claim ledger when the domain or search strategy changes.

Research Questions I

1. **Coverage and retrieval.** Which engines, queries, filters, and identifiers contributed records to each phase, and where are the known coverage limits?
2. **Structure.** How do subfields, topics, authors, venues, and citation links vary across the configured phases?
3. **Evidence.** Which claims are supported by source-backed records, which remain pending, and which are only properties of the synthetic fixture?
4. **Reproducibility.** Can a clean run regenerate the data, figures, reports, and manuscript without unexplained differences?

Scope and contribution I

The project contributes a reusable workflow rather than a domain conclusion. Retrieval and de-duplication preserve source observations; full-text assessment records access status; bibliometrics and graph construction expose descriptive structure; and the claim ledger separates fixture, configured, and source-backed statements. Subjective editorial, scientific, and visual judgments remain explicit review items even when all deterministic gates pass.

Scope and contribution I

Every generated number in the rendered manuscript must come from an artifact listed in the manifest and be traceable to the relevant method stage. Missing, stale, or misclassified evidence is a release finding, not a value to be filled by hand.